

AUTOMATIC ROOM POWER CONTROLLER

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Many a time we forget to switch off appliances like lights, fans and air-conditioners before leaving our house. This leads to considerable wastage of electricity apart from reducing the life of the appliance.

A circuit is presented here which senses the absence of occupants in a room and automatically shuts off the power to a particular appliance, say, a tubelight or fan. It turns on the appliance again when someone enters the room.

Though some circuits are already available for the purpose, most of them are expensive, require a thorough knowledge of microcontroller programming, and are too complicated to assemble by amateurs and hobbyists. Some others are too basic in their operation and prone to malfunction.

The circuit presented here is relatively simple, inexpensive and can be assembled by anyone having some knowledge of electronics. Two decade counter CD4017 ICs along with quad op-amp LM324 and some logic gates are wired to count the number of persons going inside the room and the number of persons coming out. When the numbers are equal, the circuit assumes that there is nobody inside the room and shuts the power off. As long as numbers differ, the power remains on.

Circuit description

Fig. 1 shows the circuit of the automatic room power controller. Two laser diodes and two light-dependent resistors (LDRs) are used as sensors at the entry door. This arrangement

can be replaced with the combination of IR LEDs and IR sensors or even normal bright LEDs if laser diodes are unavailable. However, laser diodes with pointed directional light give the circuit better sensitivity and range. The sensors can be placed wide apart, which makes the circuit suitable for covering a large area such as a corridor or any other particular portion of a building. Commonly available laser pointers or laser guns can also be used as laser diodes.

To derive the power supply for the circuit, the 230V, 50Hz AC mains

is stepped down by transformer X1 to deliver a secondary output of 12V-0-12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D15 and D16, filtered by capacitors C10 and C11 and regulated by IC 7809 (IC2). Capacitors C8 and C9 bypass the ripples present in the regulated supply. Regulated 9V obtained is used to power the circuit except relay RL1.

Light-dependent resistors LDR1 and LDR2 are fitted on one pillar (left or right) of the door such that one LDR is inside the door and the other outside the door. Laser diodes LD1 and LD2 are fitted on the opposite pillar of the door—one inside and the other outside—such that the light from the laser diode falls directly on the LDR placed on that side (see Fig. 2). The LDRs are housed in suitable cabinets to protect from ambient light.

When someone passes through the door, the light beams falling on the LDRs from laser diodes are interrupted one after another, making either IC3 or IC4 to count one depending on the direction of movement. Let us see how: resistor R4 and LDR1, and resistor R5 and LDR2 form voltage dividers for the 6.8V supply derived from the combination of resistor R3 and zener diode ZD1 (6.8V zener diode). The divided voltages so obtained are fed to pins 10 and 12, and pins 5 and 3 of IC1, respectively. The resistance of the LDR in bright light is less than 5 kilo-ohms, while the resistance in dark is greater than 1 mega-ohm. So the voltage at non-inverting pins 10 and 12, and pins 5 and 3 depends on the amount of light falling on the LDRs.



PARTS LIST

Semiconductors:

IC1	- LM324 quad operational amplifier
IC2	- 7809, 9V regulator
IC3, IC4	- CD4017 decade counter
IC5-IC7	- CD4081 quad AND gate
T1-T3	- BC548 npn transistor
D1-D14	- 1N4148 switching diode
D15-D18	- 1N4007 rectifier diode
ZD1	- 6.8V zener diode
ZD2	- 3.3V zener diode
LD1, LD2	- Laser diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

R1, R2	- 150-ohm
R3, R6	- 2.2-kilo-ohm
R4, R5, R14	- 10-kilo-ohm
R15, R18, R19	- 22-kilo-ohm
R7-R10, R16	- 470-kilo-ohm
R11, R12, R17	- 220-kilo-ohm
R13	- 10-kilo-ohm
LDR1, LDR2	- Light-dependent resistor

Capacitors:

C1, C2, C4, C5	- 4.7 μ F, 25V electrolytic
C6, C7	- 0.47 μ F, 25V electrolytic
C9	- 100 μ F, 25V electrolytic
C10	- 470 μ F, 35V electrolytic
C3, C8, C11	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V-0-12V, 500mA secondary transformer
RL1	- 12V, 1C/O relay
S1	- Push-to-on tactile switch